WEB DEVELOPEMENT COURSWORK REPORT

Unit Code: 6G4Z0024 – Web Development

Assignment Title: A Structured Website

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1. Usability and Accessibility Checklist

Below is list of the implemented features that enhances usability and accessibility of website, along with its brief explanation concluding how it improves the user experience.

1.1 Sementic HTML Structures

The usability of sementic elements that are: <main>; <nav>, <section>, and <header> structure the html pages as it improves the appearence and accessibility for screen readers allowing them to understand layout more accurately and helping visually impaired users to navigate site easily.

1.2 Alt Text For Images

All book covers that are optimised in the website has its name written in ALT attribute on HTML File. As it will ensure that users of screen reader or users that has slow internet connection tendencies can still understand what image represents.

1.3 Responsive Design

To make the website accessible in all devices, I implemented responsive design by using CSS techniques that are: flexbox, css grid and media queries. It ensures that website adapts to different screen size. In this case, on devices that has width of 1250px or more makes books appear in two columns but devices that width is 750px or less makes books appear only in one column. And, this improves usability and availibility on smartphones and tablets.

1.4 Mobile Friendly Navigation Menu

Adding the hamburgur icon as menu that will appear in small screens will prevent the navigation bar from becoming cluttered and will ensure user can access all pages easily on mobile devices.

1.5 Clear Form Validation Feedback

Payment form has includes JS validation that counter verifies the card number, expiry date and cvv, checks for error and display the clear messages below the form to help users to correct their mistakes and proceed payment query.

1.6 Keyboard Accessibility

All the interactive elements of website like link, buttons, form-fields are accessible via keyboard. In this case, it improves user experience and overall accessibility.

1.7 High Colour Contrast

The choice of high-contrast colours over website ensures readability and benefits users with visual impairments or low contrast sensitivity.

1.8 User Feedback on Success Page

As soon as user has made payment, the website will take the user to success payment page that will shown him/her last 4 digit of their card. In this case, the purpose of providing this to user is to reassures them that their details were proceed correctly and enhances their trust in the interface.

2. Code Explanation

This section of report will cover the key part of Javascript code and how it fulfils the assignment requirements.

2.1 Mastercard Validation Logic

For card number validation, this code will ensure that card number contains exactly 16 digits:

```
if (!/^\d{16}$/.test(cardNumber)) {

message.textContent = 'Card number must be exactly 16 digits.';

return;}
```

For master card prefix check (51-55), this code ensures that card number begins with a valid Master card prefix:

```
if (!/^(51|52|53|54|55)/.test(cardNumber)) {

message.textContent = 'Card must be a valid Mastercard (starts with 51–55).';

return;}
```

By implementing these js code will ensure that user enters a valid Mastercard number before the form is submitted.

2.2 Expiry Date Validation

I ensure that user enters the correct expiry month and year by comparing with current date. And, it will prevent user from submitting expired cards:

```
if (expYear < thisYear || (expYear === thisYear && expMonth < thisMonth)) {

message.textContent = 'The card has expired.';

return;}
```

2.3 CVV Validation

To check the cvv is correct, and it must be 3-4 digits. I implemented restriction using JS Regex that ensures the security code is valid:

```
if (!/^\d{3,4}$/.test(cvv)) {

message.textContent = 'CVV must be 3 or 4 digits.';

return;}
```

2.4 API Post Request

Once validation is bypassed, the inputed card information is passed to the server in JSON format:

```
const response = await fetch(

'https://mudfoot.doc.stu.mmu.ac.uk/node/api/creditcard',

{

method: 'POST',

headers: { 'Content-Type': 'application/json' },

body: JSON.stringify(payload)

});
```

And below code will also handle the server errors and network failures:

```
if (!response.ok) {

message.textContent = data.error || 'Payment failed. Please try again.';}
```

By doing that, this meets the requirement to validate the card both client-side and server-side.

2.5 Saving Last Four Digits for Success Page

The last four digits are stored using sessionStorage:

```
const last4 = card.slice(-4);

sessionStorage.setItem("last4", last4);
```

Then it is retrieve and displayed on Success.html:

```
document.getElementById("last4").textContent = last4 || "????";
```

This matches the requirement to show the masked card number on the success page.

2.6 Auto-Formatting Card Number

To improve readability, the spaces are inserted by default after every 4 digits:

```
value = value.replace(/(.{4})/g, '$1 ').trim();
```

2.7 Responsive Navigation Menu

I implemented a hamburger menu for mobile screens, which improves overall website usability:

```
menubar.addEventListener('click', () => {

navLinks.classList.toggle('active');
```

```
});
```